

Conditional Expectation

The lecture is organised around one picture, the table of outcomes of two dice: Y the first, Z the second, $X = Y + Z$. The rows of the table are the events $\{Y = y\}$.

1. **Given a value:** $\mathbb{E}[X \mid Y = y]$ The average of X along row y . A number for each y , hence a function of y .
2. **Law of total expectation** The average of the row averages, weighted by row probabilities, is the overall average.
3. **Given the variable:** $\mathbb{E}[X \mid Y]$ Write each row's average into every cell of that row. The result is a random variable.
4. **Properties** Tower, taking out what is known, independence. Each is a statement about the rows of the table.
5. **Two applications** Random sums, and waiting times. Both are computed by conditioning on the first stage.

Three Problems We Cannot Yet Do

Each asks for an expected value. With the tools so far, the only route is to find the distribution of the quantity first.

Problem A: a random number of coins

Roll a die, getting N . Toss N fair coins; H is the number of heads. Find $\mathbb{E}[H]$.

Via the pmf of H : $p_H(h) = \sum_{n=\max(h,1)}^6 \frac{1}{6} \binom{n}{h} 2^{-n}$ for $h = 0, \dots, 6$, seven sums, then $\sum_h h p_H(h)$.

Problem B: an insurer's payout

An insurer receives N claims in a year, $\mathbb{E}[N] = 20$, each of mean size £1500, sizes independent of the count. Find the expected total payout $S = X_1 + \dots + X_N$.

Via the distribution of S : it is a mixture over n of n -fold convolutions of the claim-size distribution, and neither distribution is given.

Problem C: waiting for HH

Toss a fair coin until two heads in a row appear; T is the number of tosses used. Find $\mathbb{E}[T]$.

Via the pmf of T : $\mathbb{P}(T = n) = F_{n-1}/2^n$ with $F_1, F_2, F_3, \dots = 1, 1, 2, 3, 5, \dots$ the Fibonacci numbers, then $\sum_n n \mathbb{P}(T = n)$.

All three are solved below by conditioning on the first stage (slides 12, 32, 35).

The Table

Two fair dice: $(\Omega, \mathcal{F}, \mathbb{P})$ with $\Omega = \{1, \dots, 6\}^2$, $\mathcal{F}$ all subsets of Ω , and $\mathbb{P}(\{\omega\}) = \frac{1}{36}$ for each outcome $\omega = (y, z)$.

- $Y: \Omega \rightarrow \{1, \dots, 6\}$, $Y(\omega) = y$, the first die: the **row**.
- $Z: \Omega \rightarrow \{1, \dots, 6\}$, $Z(\omega) = z$, the second die: the **column**.
- $X: \Omega \rightarrow \{2, \dots, 12\}$, $X(\omega) = y + z$, the sum: the **number in the cell**.

Shorthand, used throughout:

$\{Y = y\} := \{\omega \in \Omega : Y(\omega) = y\}$ and
 $\mathbb{P}(Y = y) := \mathbb{P}(\{Y = y\})$; likewise $\{X \in A\}$,
 $\mathbb{P}(X \leq x)$, $\mathbb{P}(X = x, Y = y)$.

Every random variable on this Ω assigns a number to each of the 36 cells; every event is a set of cells, with probability the number of cells over 36.

$$X(\omega) = y + z$$

$z =$	1	2	3	4	5	6
$y=1$	2	3	4	5	6	7
$y=2$	3	4	5	6	7	8
$y=3$	4	5	6	7	8	9
$y=4$	5	6	7	8	9	10
$y=5$	6	7	8	9	10	11
$y=6$	7	8	9	10	11	12

Conditioning on $Y = y$ Is Restricting to a Row

The event $\{Y = 4\}$ is the fourth row: six cells, probability $\frac{6}{36} = \frac{1}{6}$. Given $Y = 4$, the outcome lies in that row, and each of its six cells has conditional probability

$$\mathbb{P}(\{\omega\} \mid Y = 4) = \frac{\mathbb{P}(\{\omega\} \cap \{Y = 4\})}{\mathbb{P}(Y = 4)} = \frac{1/36}{1/6} = \frac{1}{6}.$$

So, given $Y = 4$, X takes each of the values 5, 6, ..., 10 with probability $\frac{1}{6}$, which is the distribution of the numbers in row 4.

row 4 is the event $\{Y = 4\}$

$z =$	1	2	3	4	5	6
$y=1$	2	3	4	5	6	7
$y=2$	3	4	5	6	7	8
$y=3$	4	5	6	7	8	9
$y=4$	5	6	7	8	9	10
$y=5$	6	7	8	9	10	11
$y=6$	7	8	9	10	11	12

Definition (conditional pmf)

Let $S_Y = \{y : p_Y(y) > 0\}$. The **conditional pmf of X given Y** is

$$p_{X|Y} : \mathbb{R} \times S_Y \rightarrow [0, 1], \quad p_{X|Y}(x \mid y) = \mathbb{P}(X = x \mid Y = y) = \frac{p_{X,Y}(x, y)}{p_Y(y)};$$

the first argument x is a value of X , the second argument y names the row.

Restrict to row y and rescale its probabilities so that they total 1.

Conditional Expectation Given $Y = y$: The Row Average

Definition (discrete case)

Let $S_Y = \{y : p_Y(y) > 0\}$. The **conditional expectation of X given $Y = y$** is the function

$$g: S_Y \rightarrow \mathbb{R}, \quad g(y) = \mathbb{E}[X \mid Y = y] = \sum_x x p_{X|Y}(x \mid y).$$

The ordinary expected value, computed with the conditional pmf of row y in place of the pmf of X .

Reading the definition

- The variable x runs over the values of X and is summed out; y is held fixed and names the row; the result depends on y alone.
- $g: S_Y \rightarrow \mathbb{R}$ is a function on the values of Y , not on Ω , so it is not a random variable.
- For $y \notin S_Y$, $g(y)$ is undefined, since a row of probability 0 has no conditional distribution.
- $\mathbb{E}[X \mathbf{1}_{\{Y=y\}}] = \sum_x x p_{X,Y}(x, y) = \mathbb{E}[X \mid Y = y] \mathbb{P}(Y = y)$, with $\mathbf{1}_{\{Y=y\}}: \Omega \rightarrow \{0, 1\}$ the indicator of the row. This is the total of the row, not its average.

Two Dice: The Row Average as a Function of y

$\mathbb{E}[X \mid Y = y]$ for the sum of two dice

Row y contains $y + 1, \dots, y + 6$, each with conditional probability $\frac{1}{6}$:

$$\mathbb{E}[X \mid Y = y] = \frac{1}{6}((y + 1) + \dots + (y + 6)) = y + 3.5.$$

As a function on $S_Y = \{1, \dots, 6\}$:

y	1	2	3	4	5	6
$g(y) = \mathbb{E}[X \mid Y = y]$	4.5	5.5	6.5	7.5	8.5	9.5

each row, then its average

$y=1$	2	3	4	5	6	7	$\rightarrow 4.5$
$y=2$	3	4	5	6	7	8	$\rightarrow 5.5$
$y=3$	4	5	6	7	8	9	$\rightarrow 6.5$
$y=4$	5	6	7	8	9	10	$\rightarrow 7.5$
$y=5$	6	7	8	9	10	11	$\rightarrow 8.5$
$y=6$	7	8	9	10	11	12	$\rightarrow 9.5$

- $g: \{1, \dots, 6\} \rightarrow \mathbb{R}$, $g(y) = y + 3.5$, takes six values, one per row.
- $g(y) - y = 3.5 = \mathbb{E}[Z]$ for every y , because on row y the first die is fixed at y and only the second die varies. Slide 9 develops this into a method.

The Continuous Case: The Row $\{Y = y\}$ Becomes a Line

Definition (continuous case)

Let X, Y have joint pdf $f_{X,Y}: \mathbb{R}^2 \rightarrow [0, \infty)$, let $S_Y = \{y : f_Y(y) > 0\}$, and define the **conditional pdf of X given Y**

$$f_{X|Y}: \mathbb{R} \times S_Y \rightarrow [0, \infty), \quad f_{X|Y}(x | y) = \frac{f_{X,Y}(x, y)}{f_Y(y)},$$

where x is a value of X and y is the observed value of Y . The **conditional expectation of X given $Y = y$** is the function

$$g: S_Y \rightarrow \mathbb{R}, \quad g(y) = \mathbb{E}[X | Y = y] = \int_{-\infty}^{\infty} x f_{X|Y}(x | y) dx.$$

The row $\{Y = y\}$ is now the horizontal line at height y , and the function $x \mapsto f_{X,Y}(x, y)/f_Y(y)$, which integrates to 1, plays the part of the row's conditional pmf.

- For fixed $y \in S_Y$, $f_{X|Y}(\cdot | y)$ is a pdf in x : $\int f_{X|Y}(x | y) dx = f_Y(y)/f_Y(y) = 1$.
- $\mathbb{P}(Y = y) = 0$, so $\mathbb{P}(\cdot | Y = y)$ cannot be defined by dividing probabilities; the ratio of densities is the limit of conditioning on $\{y \leq Y \leq y + \delta\}$ as $\delta \rightarrow 0$.
- The procedure is otherwise unchanged: fix y , divide $f_{X,Y}(\cdot, y)$ by $f_Y(y)$, and integrate x against the result.

The Two-Stage Point: Conditioning on $Y = y$ and on $X = x$

Y uniform on $[0, 1]$; given $Y = y$, X uniform on $[0, y]$. So $f_Y = 1$ on $(0, 1]$, $f_{X|Y}(x | y) = \frac{1}{y}$ on $[0, y]$, and $f_{X,Y}(x, y) = f_{X|Y}(x | y)f_Y(y) = \frac{1}{y}$ on the triangle $0 < x < y < 1$.

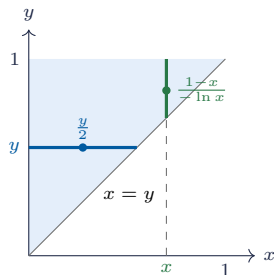
Given $Y = y$: the horizontal line at height y

$S_Y = (0, 1]$, $f_{X|Y}(\cdot | y)$ is uniform on $[0, y]$, and $\mathbb{E}[X | Y = y] = \int_0^y x \frac{1}{y} dx = \frac{y}{2}$.

Given $X = x$: the vertical line at x

$f_X(x) = \int_x^1 \frac{1}{y} dy = -\ln x$ on $S_X = (0, 1)$, so $f_{Y|X}(y | x) = \frac{1/y}{-\ln x}$ for $x < y < 1$ and $\mathbb{E}[Y | X = x] = \int_x^1 \frac{y}{y} \frac{dy}{-\ln x} = \frac{1-x}{-\ln x}$.

- Given $X = x$ the density is not uniform: $f_{Y|X}(y | x) \propto \frac{1}{y}$ gives more weight to small y , so $\mathbb{E}[Y | X = x]$ lies below the midpoint $\frac{1+x}{2}$ of $[x, 1]$; at $x = \frac{1}{2}$ it is 0.7213 against 0.75.



Blue: $\{Y = y\}$, uniform on $[0, y]$.
Green: $\{X = x\}$, density $\propto \frac{1}{y}$ on $[x, 1]$.

Functions of X , and What Is Constant on a Row

The definition applies to any random variable, and in particular to any function of X : for $W: \Omega \rightarrow \mathbb{R}$ and $h: \mathbb{R} \rightarrow \mathbb{R}$,

$$\mathbb{E}[W \mid Y = y] = \sum_w w p_{W|Y}(w \mid y), \quad \mathbb{E}[h(X) \mid Y = y] = \sum_x h(x) p_{X|Y}(x \mid y),$$

with $p_{W|Y}$ defined as on slide 4 with W in place of X ; the second formula is the rule $\mathbb{E}[h(X)] = \sum_x h(x)p_X(x)$ applied to the row's pmf.

Observation: on row y , Y is the constant y

Any random variable that depends on the outcome only through Y is constant on the row. Given $Y = y$, $X = Y + Z$ equals $y + Z$, and on the row only Z varies.

Two dice: $\mathbb{E}[X^2 \mid Y = y]$ without listing the row

On row y , $X^2 = (y + Z)^2 = y^2 + 2yZ + Z^2$ with y a constant and Z uniform on $\{1, \dots, 6\}$, so

$$\mathbb{E}[X^2 \mid Y = y] = y^2 + 2y\mathbb{E}[Z] + \mathbb{E}[Z^2] = y^2 + 7y + \frac{91}{6}.$$

Computed directly: square and average the six numbers in each of the six rows.

- Slide 21 generalises this observation as the property “taking out what is known”.

Total Expectation: The Average of the Row Averages

The rows partition the table. The overall average of the 36 cells can be computed row by row:

$$\mathbb{E}[X] = \sum_{y=1}^6 \underbrace{\frac{1}{6}}_{\mathbb{P}(Y=y)} \cdot \underbrace{(y + 3.5)}_{\text{row average}} = 7.$$

2	3	4	5	6	7	→ 4.5
3	4	5	6	7	8	→ 5.5
4	5	6	7	8	9	→ 6.5
5	6	7	8	9	10	→ 7.5
6	7	8	9	10	11	→ 8.5
7	8	9	10	11	12	→ 9.5

$$\frac{1}{6} \sum \downarrow = 7$$

Theorem (law of total expectation)

discrete:
$$\mathbb{E}[X] = \sum_{y \in S_Y} \mathbb{E}[X \mid Y = y] p_Y(y),$$

continuous:
$$\mathbb{E}[X] = \int_{S_Y} \mathbb{E}[X \mid Y = y] f_Y(y) dy.$$

Average each row, then average the row averages weighted by the row probabilities; because the rows partition the table, this is the overall average.

Total Expectation: Proof

Proof (discrete case)

$$\begin{aligned}\sum_y \mathbb{E}[X \mid Y = y] p_Y(y) &= \sum_y \left(\sum_x x p_{X|Y}(x \mid y) \right) p_Y(y) && \text{definition} \\ &= \sum_y \sum_x x p_{X,Y}(x, y) && \text{undo the rescaling} \\ &= \sum_x x \sum_y p_{X,Y}(x, y) = \sum_x x p_X(x) = \mathbb{E}[X]. && \text{swap sums; marginal} \quad \square\end{aligned}$$

Line two undoes the rescaling that defined the conditional pmf, and line three sums the cells column-first instead of row-first.

- On the finite table the two sums may be swapped freely; in general the swap requires $\mathbb{E}|X| < \infty$.
- The continuous case is the same argument with f in place of p and $\int$ in place of $\sum$:
$$\int_{S_Y} \left(\int x f_{X|Y}(x \mid y) dx \right) f_Y(y) dy = \iint x f_{X,Y}(x, y) dx dy = \int x f_X(x) dx.$$
- The same holds for any $W: \Omega \rightarrow \mathbb{R}$ in place of X : $\mathbb{E}[W] = \sum_y \mathbb{E}[W \mid Y = y] p_Y(y)$, in particular for $W = h(X)$.

Total Expectation at Work (1): Problem A, a Random Number of Coins

Roll a die, getting $N: \Omega \rightarrow \{1, \dots, 6\}$. Toss N fair coins, independent of the die; $H: \Omega \rightarrow \{0, \dots, 6\}$ is the number of heads. Find $\mathbb{E}[H]$.

By conditioning

Given $N = n$, H is binomial with n tosses (the coins are independent of the die, so the row $\{N = n\}$ does not change their distribution), so $\mathbb{E}[H \mid N = n] = \frac{n}{2}$. Then

$$\mathbb{E}[H] = \sum_{n=1}^6 \frac{n}{2} \cdot \frac{1}{6} = \frac{1}{2} \mathbb{E}[N] = 1.75.$$

Via the pmf of H

Without conditioning, one first finds $p_H(h) = \sum_{n=\max(h,1)}^6 \frac{1}{6} \binom{n}{h} 2^{-n}$ for $h = 0, \dots, 6$, seven sums (for instance $p_H(0) = \frac{63}{384}$, $p_H(1) = \frac{120}{384}$), and then computes $\sum_h h p_H(h)$.

- On the row $\{N = n\}$ the problem reduces to the mean of a binomial distribution. The distribution of H , a mixture of six binomials, is not needed.

Total Expectation at Work (2): The Two-Stage Point

Y uniform on $[0, 1]$; given $Y = y$, X uniform on $[0, y]$. Find $\mathbb{E}[X]$.

By conditioning

$\mathbb{E}[X \mid Y = y] = \frac{y}{2}$ and $f_Y = 1$ on $(0, 1]$, so $\mathbb{E}[X] = \int_0^1 \frac{y}{2} \cdot 1 \, dy = \frac{1}{4}$.

Via the pdf of X

Without conditioning, one first finds the pdf of X by integrating out y :

$f_X(x) = \int_x^1 f_{X,Y}(x, y) \, dy = \int_x^1 \frac{dy}{y} = -\ln x$ for $0 < x < 1$. Then $\mathbb{E}[X] = \int_0^1 x(-\ln x) \, dx$, which is evaluated by integration by parts and equals $\frac{1}{4}$.

Total Expectation at Work (3): Variance

The two-stage point again: Y uniform on $[0, 1]$; given $Y = y$, X uniform on $[0, y]$. Find $\text{Var}(X)$.

Definition (variance)

For $X: \Omega \rightarrow \mathbb{R}$ with $\mathbb{E}[X^2] < \infty$: $\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$.

The mean squared distance of X from its mean; computing it requires $\mathbb{E}[X^2]$, which can be found by total expectation.

- Here $\mathbb{E}[X^2 \mid Y = y] = \frac{y^2}{3}$, so $\mathbb{E}[X^2] = \int_0^1 \frac{y^2}{3} dy = \frac{1}{9}$ and $\text{Var}(X) = \frac{1}{9} - \frac{1}{16} = \frac{7}{144}$. Neither computation uses f_X .

$\mathbb{E}[X \mid Y]$: Every Cell of Row y Takes the Value $\mathbb{E}[X \mid Y = y]$

Slide 6 gave a function on the values of Y :

$$g: S_Y \rightarrow \mathbb{R}, g(y) = \mathbb{E}[X \mid Y = y].$$

Now write $g(y)$ into *every cell of row y* . The result is a new number in each cell of the table, that is, a new random variable.

Definition

The **conditional expectation of X given Y** is the random variable

$$\mathbb{E}[X \mid Y]: \Omega \rightarrow \mathbb{R},$$

$$\mathbb{E}[X \mid Y](\omega) = g(Y(\omega)) = \mathbb{E}[X \mid Y = Y(\omega)].$$

The value at ω is the average of the row that contains ω .

- $g(Y)$ is defined on the event $\{Y \in S_Y\}$, which has probability 1; set it to 0 elsewhere. On the two-dice Ω every outcome has positive probability, so this does not arise.

$$\mathbb{E}[X \mid Y](\omega) = Y(\omega) + 3.5$$

$z =$	1	2	3	4	5	6
$y=1$	4.5	4.5	4.5	4.5	4.5	4.5
$y=2$	5.5	5.5	5.5	5.5	5.5	5.5
$y=3$	6.5	6.5	6.5	6.5	6.5	6.5
$y=4$	7.5	7.5	7.5	7.5	7.5	7.5
$y=5$	8.5	8.5	8.5	8.5	8.5	8.5
$y=6$	9.5	9.5	9.5	9.5	9.5	9.5

Three Tables, Three Levels of Knowledge

	X					
$z=$	1	2	3	4	5	6
$y=1$	2	3	4	5	6	7
$y=2$	3	4	5	6	7	8
$y=3$	4	5	6	7	8	9
$y=4$	5	6	7	8	9	10
$y=5$	6	7	8	9	10	11
$y=6$	7	8	9	10	11	12

the outcome ω is known

	$\mathbb{E}[X Y]$					
$z=$	1	2	3	4	5	6
$y=1$	4.5	4.5	4.5	4.5	4.5	4.5
$y=2$	5.5	5.5	5.5	5.5	5.5	5.5
$y=3$	6.5	6.5	6.5	6.5	6.5	6.5
$y=4$	7.5	7.5	7.5	7.5	7.5	7.5
$y=5$	8.5	8.5	8.5	8.5	8.5	8.5
$y=6$	9.5	9.5	9.5	9.5	9.5	9.5

only the row $Y(\omega)$ is known

	$\mathbb{E}[X]$					
$z=$	1	2	3	4	5	6
$y=1$	7	7	7	7	7	7
$y=2$	7	7	7	7	7	7
$y=3$	7	7	7	7	7	7
$y=4$	7	7	7	7	7	7
$y=5$	7	7	7	7	7	7
$y=6$	7	7	7	7	7	7

neither is known

Each table is obtained from the one to its left by averaging over what is not known.

- $\mathbb{E}[X | Y = y]$: a number for each y . $\mathbb{E}[X | Y] = g(Y)$: a random variable, constant on each row $\{Y = y\}$, where its value is $\mathbb{E}[X | Y = y]$.

$\mathbb{E}[X \mid Y]$: Extreme Cases and the Continuous Case

- $\mathbb{E}[X \mid X] = X$. Conditioning on X itself, each “row” $\{X = x\}$ contains only cells where $X = x$, so its average is x .
- $\mathbb{E}[X \mid Y] = \mathbb{E}[X]$ if Y is constant. There is then a single row, the table itself, and its average is $\mathbb{E}[X]$.
- $\mathbb{E}[Y \mid Y] = Y$, and more generally $\mathbb{E}[h(Y) \mid Y] = h(Y)$ for $h: \mathbb{R} \rightarrow \mathbb{R}$: on row y , $h(Y)$ is the constant $h(y)$, whose average is $h(y)$.

Continuous case

The same definition, $\mathbb{E}[X \mid Y] = g(Y)$ with $g: S_Y \rightarrow \mathbb{R}$, $g(y) = \int x f_{X|Y}(x \mid y) dx$. The “rows” $\{Y = y\}$ each have probability 0, but $g(Y): \Omega \rightarrow \mathbb{R}$ is an ordinary random variable, namely a function of Y .

Examples

Two-stage point: $g(y) = y/2$, so $\mathbb{E}[X \mid Y] = Y/2$. At an outcome where $Y = 0.8$, $\mathbb{E}[X \mid Y]$ takes the value 0.4, whatever the value of X . Likewise $\mathbb{E}[Y \mid X] = (1 - X)/(-\ln X)$.

Random coins: $\mathbb{E}[H \mid N] = N/2$, a random variable with values $\frac{1}{2}, 1, \dots, 3$, each with probability $\frac{1}{6}$.

Property 1: Tower. The Table of Row Averages Has the Same Average

Replacing every entry of a row by the row average does not change the row's total. So the table of row averages has the same overall average as the original:

Theorem (tower property)

$$\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X].$$

Proof

$\mathbb{E}[X \mid Y] = g(Y)$, and the expectation of a function of Y is $\mathbb{E}[g(Y)] = \sum_y g(y) p_Y(y) = \sum_y \mathbb{E}[X \mid Y = y] p_Y(y) = \mathbb{E}[X]$ by total expectation. $\square$

This is the law of total expectation, with the sum over y written as the expectation of a function of Y .

- Since $\mathbb{E}[X \mid Y]$ is itself a random variable, it can in turn be conditioned on another variable.

2	3	4	5	6	7
3	4	5	6	7	8
4	5	6	7	8	9
5	6	7	8	9	10
6	7	8	9	10	11
7	8	9	10	11	12

average 7



4.5	4.5	4.5	4.5	4.5	4.5
5.5	5.5	5.5	5.5	5.5	5.5
6.5	6.5	6.5	6.5	6.5	6.5
7.5	7.5	7.5	7.5	7.5	7.5
8.5	8.5	8.5	8.5	8.5	8.5
9.5	9.5	9.5	9.5	9.5	9.5

average 7

Property 1: Why the Tower Matters

To compute $\mathbb{E}[X]$, compute $\mathbb{E}[X \mid Y]$ for a suitable Y and take its expectation.

problem	$\mathbb{E}[X \mid Y]$, a formula in Y	then $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]]$
coins, N from a die	$\mathbb{E}[H \mid N] = N/2$	$\mathbb{E}[N]/2 = 1.75$
two-stage point	$\mathbb{E}[X \mid Y] = Y/2$	$\mathbb{E}[Y]/2 = 1/4$
random sum $S = X_1 + \cdots + X_N$	$\mathbb{E}[S \mid N] = \mu N$	$\mu \mathbb{E}[N]$
waiting time T for HH	$\mathbb{E}[T \mid Y]$ for the opening Y	an equation for $\mathbb{E}[T]$

- In each case the distribution of X is laborious to find, while $\mathbb{E}[X \mid Y]$ is a simple function of Y .
- The method depends on the choice of Y : on each row the problem must be one that can be solved directly, and p_Y or f_Y must be known.
- The last two rows are the applications at the end of the lecture.

Property 2: A Problem. The Covariance of N and H

Problem A again: N the die, H the number of heads among N coins. More coins should mean more heads on average; the covariance measures the strength of this dependence.

Definition (covariance)

For $X, Y: \Omega \rightarrow \mathbb{R}$ with finite second moments:

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]; \quad \text{Cov}(X, X) = \text{Var}(X).$$

Positive when X above its mean tends to come with Y above its mean; zero for independent X, Y .

Direct computation

$\mathbb{E}[NH] = \sum_{n=1}^6 \sum_{h=0}^n n h p_{N,H}(n, h)$ with $p_{N,H}(n, h) = \frac{1}{6} \binom{n}{h} 2^{-n}$, a sum over the 21 nonzero terms of the joint pmf.

By conditioning

By the tower property, $\mathbb{E}[NH] = \mathbb{E}[\mathbb{E}[NH | N]]$. On the row $\{N = n\}$ the factor N equals the number n . If a constant factor can be taken out of a row average, then $\mathbb{E}[NH | N = n] = n \mathbb{E}[H | N = n] = n \cdot \frac{n}{2}$, and $\mathbb{E}[NH]$ becomes the expectation of a function of N . Slide 9 used this one row at a time; the next slide states it for all rows at once.

Property 2: Taking Out What Is Known

On row y , anything that is a function of Y is the constant $h(y)$, and a constant factor comes out of an average.

Theorem

For any $h: \mathbb{R} \rightarrow \mathbb{R}$ with $\mathbb{E}|h(Y)X| < \infty$: $\mathbb{E}[h(Y)X \mid Y] = h(Y) \mathbb{E}[X \mid Y]$. In particular $\mathbb{E}[h(Y) \mid Y] = h(Y)$.

Proof (discrete case)

Fix $y \in S_Y$.

$\mathbb{E}[h(Y)X \mid Y = y] = \sum_x h(y) x p_{X|Y}(x \mid y) = h(y) \sum_x x p_{X|Y}(x \mid y) = h(y) \mathbb{E}[X \mid Y = y]$. This holds for every y ; substitute $y = Y(\omega)$. The continuous case is the same, with f for p and $\int$ for $\sum$. $\square$

Whatever is already determined by the row is a constant on the row and can be taken out of the row average.

Linearity

$\mathbb{E}[aX + bX' \mid Y] = a \mathbb{E}[X \mid Y] + b \mathbb{E}[X' \mid Y]$ for constants a, b , because expectation is linear on each row. Combined with the theorem, $\mathbb{E}[h(Y) + X \mid Y] = h(Y) + \mathbb{E}[X \mid Y]$.

Property 2: Back to $\text{Cov}(N, H)$, and $\mathbb{E}[XY]$ for Two Dice

$\text{Cov}(N, H)$ for Problem A

$$\mathbb{E}[NH] = \mathbb{E}[\mathbb{E}[NH \mid N]] = \mathbb{E}[N \mathbb{E}[H \mid N]] = \mathbb{E}\left[N \cdot \frac{N}{2}\right] = \frac{1}{2} \mathbb{E}[N^2] = \frac{1}{2} \cdot \frac{91}{6} = \frac{91}{12},$$

so $\text{Cov}(N, H) = \frac{91}{12} - \frac{7}{2} \cdot \frac{7}{4} = \frac{35}{24} \approx 1.46$. With $\text{Var}(N) = \mathbb{E}[N^2] - \mathbb{E}[N]^2 = \frac{91}{6} - \frac{49}{4} = \frac{35}{12}$.
 $\text{Cov}(N, H) / \text{Var}(N) = \frac{1}{2}$, the slope of $\mathbb{E}[H \mid N] = \frac{1}{2}N$.

$\mathbb{E}[XY]$ for two dice, without listing 36 products

$$\mathbb{E}[XY] = \mathbb{E}[\mathbb{E}[XY \mid Y]] = \mathbb{E}[Y \mathbb{E}[X \mid Y]] = \mathbb{E}[Y(Y + 3.5)] = \mathbb{E}[Y^2] + 3.5 \mathbb{E}[Y] = \frac{91}{6} + \frac{49}{4} = \frac{329}{12},$$

about 27.42, so $\text{Cov}(X, Y) = \frac{329}{12} - 7 \cdot 3.5 = \frac{35}{12}$.

- Both computations follow the same three steps: the tower property, taking out the factor that is a function of the conditioning variable, and an ordinary expectation of a function of that variable.
- The 21-term double sum of the previous slide is not needed.

Property 3: Independence

Definition (independence)

$X, Y: \Omega \rightarrow \mathbb{R}$ are **independent**, written $X \perp\!\!\!\perp Y$, if

$$\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A) \mathbb{P}(Y \in B) \quad \text{for all } A, B \subseteq \mathbb{R}.$$

$\xi_1, \dots, \xi_n: \Omega \rightarrow \mathbb{R}$ are **independent** if

$\mathbb{P}(\xi_1 \in A_1, \dots, \xi_n \in A_n) = \prod_{i=1}^n \mathbb{P}(\xi_i \in A_i)$ for all $A_i \subseteq \mathbb{R}$; a sequence, if every finite part is.

Knowing the row gives no information about the column.

- Two dice: $p_{Y,Z}(y, z) = \frac{1}{36} = \frac{1}{6} \cdot \frac{1}{6} = p_Y(y) p_Z(z)$; summing over the cells of $\{Y \in A\} \cap \{Z \in B\}$ gives the definition for all A, B . So $Y \perp\!\!\!\perp Z$.

Fact (stated without proof)

Functions of disjoint groups of independent variables are independent: $\xi_1, \xi_2, \dots$ independent implies $f(\xi_1, \dots, \xi_5) \perp\!\!\!\perp g(\xi_6, \xi_7, \xi_8)$ for all $f: \mathbb{R}^5 \rightarrow \mathbb{R}$, $g: \mathbb{R}^3 \rightarrow \mathbb{R}$.

$Z(\omega) = z$, the same in every row

$z =$	1	2	3	4	5	6
$y=1$	1	2	3	4	5	6
$y=2$	1	2	3	4	5	6
$y=3$	1	2	3	4	5	6
$y=4$	1	2	3	4	5	6
$y=5$	1	2	3	4	5	6
$y=6$	1	2	3	4	5	6

Property 3: A Problem. A Random Walk, Seen From Halfway

Let $\xi_1, \xi_2, \dots : \Omega \rightarrow \{-1, 1\}$ be independent with $\mathbb{P}(\xi_i = 1) = \mathbb{P}(\xi_i = -1) = \frac{1}{2}$, and $S_n = \xi_1 + \dots + \xi_n$ the position after n steps. Suppose the first five steps have been observed, so that S_5 is known. Find $\mathbb{E}[S_8 \mid S_5]$ and $\mathbb{E}[S_8^2 \mid S_5]$.

Direct computation

There are six rows $\{S_5 = s\}$, $s \in \{-5, -3, -1, 1, 3, 5\}$. On row s , S_8 has the conditional pmf $p_{S_8|S_5}(t \mid s) = p_{S_5, S_8}(s, t) / p_{S_5}(s)$, which requires the joint pmf of (S_5, S_8) , obtained by counting paths. For $s = 3$, S_8 takes the values $t = 0, 2, 4, 6$ with probabilities $\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8}$, so the row has mean 3 and second moment 12. The other five rows are treated in the same way.

By conditioning

Write $S_8 = S_5 + D$ with $D = \xi_6 + \xi_7 + \xi_8$. On the row $\{S_5 = s\}$ the first term is the constant s and can be taken out. The second term is a function of steps that the row does not involve, so restricting to the row should not change its distribution, and $\mathbb{E}[D \mid S_5]$ should equal $\mathbb{E}[D] = 0$.

- The statement that the row does not involve D is the independence $D \perp\!\!\!\perp S_5$, which holds by the fact on the previous slide. The next slide states its consequence for conditional expectations.

Property 3: Independence. Every Row Has the Distribution of the Table

Theorem

If $X \perp\!\!\!\perp Y$ then $\mathbb{E}[X \mid Y] = \mathbb{E}[X]$, a constant; equivalently $\mathbb{E}[X \mid Y = y] = \mathbb{E}[X]$ for every $y \in S_Y$.

Proof (discrete case)

Take $A = \{x\}$, $B = \{y\}$ in the definition: $p_{X,Y}(x, y) = p_X(x) p_Y(y)$. So for every $y \in S_Y$,

$$p_{X|Y}(x \mid y) = \frac{p_X(x) p_Y(y)}{p_Y(y)} = p_X(x) :$$

each row carries the same distribution of X as the table itself. Hence every row average is $\sum_x x p_X(x) = \mathbb{E}[X]$. The continuous case is the same, using $f_{X,Y} = f_X f_Y$. □

Knowing the row gives no information about X , so the row average is the overall average.

- Two dice: $Z \perp\!\!\!\perp Y$; every row of the Z -table is $1, \dots, 6$, so $\mathbb{E}[Z \mid Y] = 3.5$ on every row.
- Also $\mathbb{E}[h(X) \mid Y] = \mathbb{E}[h(X)]$ for $h: \mathbb{R} \rightarrow \mathbb{R}$, since $h(X) \perp\!\!\!\perp Y$ too ($\{h(X) \in A\} = \{X \in h^{-1}(A)\}$).

Property 3: The Random Walk Done, and a Method

$$\mathbb{E}[S_8 \mid S_5] \text{ and } \mathbb{E}[S_8^2 \mid S_5]$$

$D = \xi_6 + \xi_7 + \xi_8$ takes the values $-3, -1, 1, 3$ with probabilities $\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8}$, so $\mathbb{E}[D] = 0$ and $\mathbb{E}[D^2] = 9 \cdot \frac{1}{4} + 1 \cdot \frac{3}{4} = 3$; and $D \perp\!\!\!\perp S_5$ (disjoint groups of steps).

$$\mathbb{E}[S_8 \mid S_5] = \mathbb{E}[S_5 \mid S_5] + \mathbb{E}[D \mid S_5] = S_5 + \mathbb{E}[D] = S_5,$$

$$\mathbb{E}[S_8^2 \mid S_5] = \mathbb{E}[S_5^2 + 2S_5D + D^2 \mid S_5] = S_5^2 + 2S_5 \mathbb{E}[D] + \mathbb{E}[D^2] = S_5^2 + 3.$$

Check on row $s = 3$: $\frac{1}{8} \cdot 0 + \frac{3}{8} \cdot 4 + \frac{3}{8} \cdot 16 + \frac{1}{8} \cdot 36 = 12 = 3^2 + 3$.

Method

To compute $\mathbb{E}[X \mid Y]$ without listing rows, write X as sums and products of *functions of Y* and *quantities independent of Y* . The former come out unchanged (taking out what is known); the latter are replaced by their ordinary expectations (independence).

- Two dice: $X = Y + Z$ with $Z \perp\!\!\!\perp Y$, so $\mathbb{E}[X \mid Y] = Y + \mathbb{E}[Z] = Y + 3.5$; and $\mathbb{E}[X^2 \mid Y] = Y^2 + 2Y \mathbb{E}[Z] + \mathbb{E}[Z^2]$.
- $\mathbb{E}[S_8 \mid S_5] = S_5$: the expected future position of the walk is its present position. This is the martingale property, taken up in the stochastic-processes lectures.

Property 3: What Independence Does Not Say

The converse fails: $\mathbb{E}[X | Y] = \mathbb{E}[X]$ does not imply $X \perp\!\!\!\perp Y$

Y uniform on $\{-1, 0, 1\}$ and $W = Y^2$. Conditioning on W creates two rows:

row	outcomes	values of Y	row average
$\{W = 0\}$	$\{0\}$	0	0
$\{W = 1\}$	$\{-1, 1\}$	-1, 1	0

So $\mathbb{E}[Y | W] = 0 = \mathbb{E}[Y]$, yet W is a function of Y :

$$\mathbb{P}(Y = 0, W = 0) = \frac{1}{3} \neq \mathbb{P}(Y = 0)\mathbb{P}(W = 0) = \frac{1}{9}.$$

- Equal row *averages* are a weaker condition than equal row *distributions*. Independence is the statement that the row distributions are equal; the theorem uses only equality of the row averages, and $\mathbb{E}[X | Y] = \mathbb{E}[X]$ asserts only that.
- $\mathbb{E}[X | Y] = \mathbb{E}[X]$ does imply, by the tower property and taking out what is known, that $\mathbb{E}[h(Y)X] = \mathbb{E}[h(Y)\mathbb{E}[X | Y]] = \mathbb{E}[h(Y)]\mathbb{E}[X]$, that is, $\text{Cov}(h(Y), X) = 0$ for every $h: \mathbb{R} \rightarrow \mathbb{R}$. In the example, $\text{Cov}(W, Y) = \mathbb{E}[Y^3] - \mathbb{E}[W]\mathbb{E}[Y] = 0$.

Properties: Summary

property	statement	reason
tower	$\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X]$	replacing each row by its average keeps the total
linearity	$\mathbb{E}[aX + bX' \mid Y] = a\mathbb{E}[X \mid Y] + b\mathbb{E}[X' \mid Y]$	averages are linear, row by row
taking out what is known	$\mathbb{E}[h(Y)X \mid Y] = h(Y)\mathbb{E}[X \mid Y]$	$h(Y)$ is constant on a row
known variables	$\mathbb{E}[h(Y) \mid Y] = h(Y), \quad \mathbb{E}[X \mid X] = X$	the average of a constant is that constant
independence	$X \perp\!\!\!\perp Y \Rightarrow \mathbb{E}[X \mid Y] = \mathbb{E}[X]$	every row has the table's distribution of X

object	type	meaning
$\mathbb{E}[X]$	number	average over the table
$\mathbb{E}[X \mid Y = y]$	number for each $y \in S_Y$; a function $g: S_Y \rightarrow \mathbb{R}$	average over row y
$\mathbb{E}[X \mid Y]$	random variable $g(Y): \Omega \rightarrow \mathbb{R}$	row average, written into every cell of the row

Application 1: Random Sums. Problem B

Let $N: \Omega \rightarrow \{0, 1, 2, \dots\}$ and let $X_1, X_2, \dots: \Omega \rightarrow \mathbb{R}$ have $\mathbb{E}[X_i] = \mu$ for all i . Put

$$S = X_1 + X_2 + \dots + X_N \quad (S = 0 \text{ when } N = 0).$$

There are two sources of randomness, the number of terms and the values of the terms.

Problem B, via the distribution of S

N claims, $\mathbb{E}[N] = 20$; claim i has size X_i with $\mathbb{E}[X_i] = \mathcal{L}1500$. Find $\mathbb{E}[S]$. The pmf or pdf of S is $\sum_n \mathbb{P}(N = n) \cdot (\text{distribution of } X_1 + \dots + X_n)$, involving an n -fold convolution for each n ; and here neither the distribution of N nor that of the X_i is specified, only the two means.

By conditioning

On the row $\{N = n\}$, $S = X_1 + \dots + X_n$ is a sum of a *fixed* number of terms, and $\mathbb{E}[X_1 + \dots + X_n] = n\mu$ by linearity. So $\mathbb{E}[S \mid N = n]$ would be $n\mu$, provided that restricting to the row does not change the distribution of the terms.

- This proviso is an assumption about how N is determined. It is stated on the next slide.

Application 1: Random Sums. The Assumption

Assumption: N is independent of the X_i

For every n and all sets $A_1, \dots, A_n \subseteq \mathbb{R}$,

$$\mathbb{P}(N = n, X_1 \in A_1, \dots, X_n \in A_n) = \mathbb{P}(N = n) \mathbb{P}(X_1 \in A_1, \dots, X_n \in A_n).$$

Given $N = n$, the joint distribution of $(X_1, \dots, X_n)$ is the same as the unconditional one.

- It is not assumed that the X_i are independent of each other or identically distributed, only that they have the common mean μ .
- The assumption holds when N is determined by a mechanism separate from the X_i : the number of customers is fixed before any of them spends, or the number of claims does not depend on their sizes.
- It fails when N is determined by the values of the X_i , as in “stop at the first six” or “include the second term only if it is small” (slide 32).

Application 1: Random Sums. Theorem and Proof

Theorem

Under the assumption, $\mathbb{E}[S \mid N = n] = n\mu$ for every n , hence $\mathbb{E}[S \mid N] = \mu N$ and $\mathbb{E}[S] = \mu \mathbb{E}[N]$.

Proof

On the row $\{N = n\}$, $S = W_n := X_1 + \cdots + X_n$. By the assumption, the row $\{N = n\}$ does not change the distribution of W_n , so Property 3, applied to W_n on the row $\{N = n\}$, gives

$$\mathbb{E}[S \mid N = n] = \mathbb{E}[W_n \mid N = n] = \mathbb{E}[W_n] = n\mu$$

by linearity. Substitute $n = N$ to get $\mathbb{E}[S \mid N] = \mu N$; then $\mathbb{E}[S] = \mathbb{E}[\mu N] = \mu \mathbb{E}[N]$ by the tower property. □

On the row $\{N = n\}$ the number of terms is fixed, and the mean of the sum follows by linearity.

- The argument has the same three steps as before: condition on N , compute the row average by linearity, and apply the tower property.
- Only the mean of S is computed, not its distribution.

Application 1: Back to the Insurer, and When the Assumption Fails

Problem B

$\mathbb{E}[S] = \mu \mathbb{E}[N] = 1500 \cdot 20 = \pounds 30\,000$, whatever the distributions of N and of the claim sizes, as long as the count is independent of the sizes.

The formula fails when N depends on a term

Two dice X_1, X_2 . Let $N = 1$ if $X_2 = 6$ and $N = 2$ otherwise (the second die counts only if it is not a six). Then $S = X_1 + X_2 \mathbf{1}_{\{X_2 \neq 6\}}$ and

$$\mathbb{E}[S] = 3.5 + \frac{1+2+3+4+5}{6} = 6, \quad \text{but} \quad \mu \mathbb{E}[N] = 3.5 \cdot \frac{11}{6} = \frac{77}{12} \approx 6.42.$$

On the row $\{N = 2\} = \{X_2 \neq 6\}$ the second term averages 3, not $\mu = 3.5$; restricting to the row changes the distribution of a term.

- “Stop at the first six” also violates the assumption, yet $\mu \mathbb{E}[N] = 3.5 \cdot 6 = 21$ is correct. This is Wald’s identity, a separate theorem, whose hypothesis is that the decision to stop after n terms depends only on the first n terms. In the example above, whether $N = 1$ is decided by the value of X_2 , a later term.

Application 2: Waiting for HH. Problem C

Toss a fair coin repeatedly. Let $T: \Omega \rightarrow \{2, 3, \dots\}$ be the number of the toss on which two consecutive heads are first completed: for THTHH..., $T = 5$. Find $\mathbb{E}[T]$.

Via the pmf of T

$\mathbb{P}(T = n)$ is the number of H/T strings of length n that end in HH and contain no earlier HH, over 2^n . The count is the Fibonacci number F_{n-1} , so $\mathbb{P}(T = n) = F_{n-1}/2^n$, and the sum $\mathbb{E}[T] = \sum_{n \geq 2} n F_{n-1}/2^n$ is evaluated with generating functions.

Here the rows are described by events rather than by the values of a named variable, which requires the following definition.

Definition (conditional expectation given an event)

For $A \in \mathcal{F}$ with $\mathbb{P}(A) > 0$ and $\mathbf{1}_A: \Omega \rightarrow \{0, 1\}$ the indicator of A ,

$$\mathbb{E}[X \mid A] = \sum_x x \mathbb{P}(X = x \mid A) = \mathbb{E}[X \mid \mathbf{1}_A = 1].$$

A row may be any event A of positive probability; restrict to A , rescale, and average.

Application 2: Waiting for HH. The Restart Property

Let A_k be the event that the first k tosses form a given H/T string ending in T; $\mathbb{P}(A_k) = 2^{-k} > 0$.
On A_k no progress towards HH has been made at toss k .

Restart property

Let T' be the number of tosses after toss k until HH is first completed among tosses $k+1, k+2, \dots$.
Then on A_k , $T = k + T'$, and

$$\mathbb{E}[T \mid A_k] = k + \mathbb{E}[T' \mid A_k] = k + \mathbb{E}[T].$$

Why

T' is a function of tosses $k+1, k+2, \dots$, and $\mathbf{1}_{A_k}$ of tosses $1, \dots, k$: disjoint groups of independent tosses, so $T' \perp\!\!\!\perp \mathbf{1}_{A_k}$ and, by Property 3, $\mathbb{E}[T' \mid A_k] = \mathbb{E}[T']$. The tosses $k+1, k+2, \dots$ form a sequence of fair coin tosses, which is the experiment defining T , so T' has the distribution of T and $\mathbb{E}[T'] = \mathbb{E}[T]$. □

On such a row, T is k plus an independent copy of T , so the row average is k plus the unknown mean.

Application 2: Waiting for HH. The Computation

The conditioning variable. Let $Y: \Omega \rightarrow \{1, 2, 3\}$ record how the game opens:

$$Y = \begin{cases} 1 & \text{toss 1 is T} \\ 2 & \text{tosses 1,2 are H,T} \\ 3 & \text{tosses 1,2 are H,H} \end{cases}$$

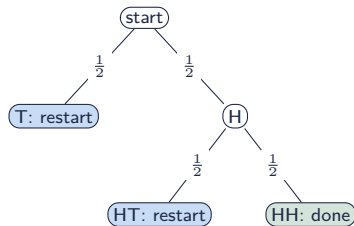
with $p_Y(1) = \frac{1}{2}$, $p_Y(2) = p_Y(3) = \frac{1}{4}$. The three cases are disjoint and cover Ω , so Y is defined at every outcome and there are three rows.

Row averages. Rows 1 and 2 are the events $A_1 = \{\text{T}\}$ and $A_2 = \{\text{HT}\}$ (restart); row 3 is finished:

$$\mathbb{E}[T \mid Y = 1] = 1 + \mathbb{E}[T], \quad \mathbb{E}[T \mid Y = 2] = 2 + \mathbb{E}[T],$$

and $\mathbb{E}[T \mid Y = 3] = 2$. **Total expectation.**

$$\begin{aligned} \mathbb{E}[T] &= \frac{1}{2}(1 + \mathbb{E}[T]) + \frac{1}{4}(2 + \mathbb{E}[T]) + \frac{1}{4} \cdot 2 \\ &= \frac{3}{4}\mathbb{E}[T] + \frac{3}{2}, \quad \text{so } \mathbb{E}[T] = 6. \end{aligned}$$



Two of the three leaves lead back to the start, which is why $\mathbb{E}[T]$ appears on both sides of the equation.

Application 2: Further Waiting Times

When a process restarts after certain outcomes, conditioning on the first step yields an equation in which the unknown mean appears on both sides, and solving it determines the mean.

- **Rolls until the first six**, T_6 : $\mathbb{E}[T_6] = \frac{1}{6} \cdot 1 + \frac{5}{6} (1 + \mathbb{E}[T_6])$, so $\mathbb{E}[T_6] = 6$.
- **Waiting for a T**, T_T : the same equation with $\frac{1}{2}$ for $\frac{1}{6}$ gives $\mathbb{E}[T_T] = 2$.
- **Pattern HT**, T_{HT} : after the first H, wait for a T. $\mathbb{E}[T_{HT}] = \frac{1}{2} (1 + \mathbb{E}[T_{HT}]) + \frac{1}{2} (1 + \mathbb{E}[T_T])$, so $\mathbb{E}[T_{HT}] = 4 < 6$. When an attempt at HT fails (with HH), the final H begins a new attempt; when an attempt at HH fails (with HT), no progress is kept.
- **Pattern HHH**: 14. **Two sixes in a row**: 42. The equations have the same form, with more rows.
- **Conditioning on three failures**: $\mathbb{E}[T_6 \mid T_6 > 3] = 3 + \mathbb{E}[T_6] = 9$, by the restart property at $k = 3$.

Finiteness of $\mathbb{E}[T]$

Solving $\mathbb{E}[T] = \frac{3}{4}\mathbb{E}[T] + \frac{3}{2}$ presumes $\mathbb{E}[T] < \infty$. Each block of two tosses is HH with probability $\frac{1}{4}$, so $\mathbb{P}(T > 2m) \leq (\frac{3}{4})^m$, and $\mathbb{E}[T] = \sum_{n \geq 0} \mathbb{P}(T > n) < \infty$ (tail-sum formula for $\{0, 1, 2, \dots\}$ -valued T ; standard, not proved here).